AI-Based Endemic Bird Occurrence and Biodiversity Prediction

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Project Proposal Report

N. Sajiththiya – IT23343870

B.Sc. (Hons) Degree in Information Technology Specialized in Information Technology

Department of Computing

Sri Lanka Institute of Information Technology Sri Lanka

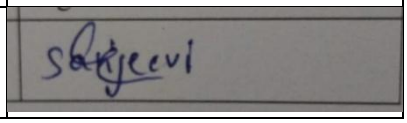
September 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature
SAJITHTHIYA.N	IT23343870	

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature
Ms. Sanjeevi Chandrasiri	14/09/2026	

ABSTRACT

Historical bird occurrence records provide valuable information about where and when a species has previously been observed. However, an individual observation does not indicate whether the same species is likely to occur in another location, during a different time period, or under different environmental conditions. This limitation is particularly important for Sri Lankan endemic birds, as reliable information about their spatial and seasonal occurrence can support biodiversity monitoring, field surveys, and conservation planning.

This research proposes an AI-based spatial-temporal approach for predicting the occurrence of selected Sri Lankan endemic bird species and identifying patterns of endemic-bird biodiversity. Historical occurrence records from eBird and GBIF will be used as the primary data sources. These records will be combined with environmental and geographic variables, including temperature, rainfall, land cover, vegetation condition, and elevation. Temporal features such as month and season, together with spatial features derived from latitude and longitude, will also be incorporated to prepare a structured dataset for machine learning.

Random Forest and XGBoost will be developed and compared for bird occurrence prediction. The models will learn relationships between previous bird observations and the associated spatial, temporal, and environmental conditions. Their performance will be evaluated using precision, recall, F1-score, and ROC-AUC, supported by spatially and temporally aware validation to reduce the influence of data leakage. SHAP and feature-importance analysis will also be used to understand the contribution of different variables to the model predictions.

The component will produce occurrence-probability maps for selected endemic bird species, endemic-bird biodiversity maps, and model-supported candidate observation areas. A location-aware feature will also calculate the distance between a user's location and suitable candidate areas. These outputs will represent predicted probabilities rather than guaranteed bird sightings. Within the complete conservation intelligence framework, Component 03 provides the occurrence and biodiversity prediction layer to support biodiversity monitoring, field survey planning, and conservation activities.

Keywords: Bird Occurrence Prediction, Biodiversity, Random Forest, XGBoost, GIS

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AUC	Area Under the Curve
DEM	Digital Elevation Model
GBIF	Global Biodiversity Information Facility
GIS	Geographic Information System
ML	Machine Learning
NDVI	Normalized Difference Vegetation Index
PR-AUC	Area Under the Precision-Recall Curve
RF	Random Forest
ROC-AUC	Receiver Operating Characteristic – Area Under the Curve
SDM	Species Distribution Model
SHAP	SHapley Additive exPlanations
SRTM	Shuttle Radar Topography Mission
XGBoost	Extreme Gradient Boosting

1. INTRODUCTION

1.1 Background and Context

Sri Lanka has a rich and distinctive bird community, with many endemic species associated with specific forest types, elevations, and environmental conditions. Understanding where these species occur is important for biodiversity research, field monitoring, and conservation planning. Historical bird occurrence records provide valuable information about previous observations; however, they mainly indicate that a particular species was recorded at a specific location and time. These records alone cannot determine whether the same species is likely to occur in another location, during a different time, or under different environmental conditions.

This presents a practical challenge for researchers and field teams. Bird observations are collected from different locations and time periods, while environmental conditions also vary across space and time. Therefore, analyzing historical observations together with environmental information can provide a better understanding of the factors associated with bird occurrence. Community-science platforms such as eBird have already been used internationally for bird distribution modelling. However, previous research has also highlighted the importance of considering factors such as spatial sampling bias, observation effort, and seasonal variation when using these data for predictive modelling [1]– [3].

To address this problem, Component 03 proposes a machine-learning-based approach for predicting the occurrence and biodiversity patterns of selected Sri Lankan endemic birds. Historical bird occurrence records will be obtained primarily from eBird and GBIF and combined with environmental, spatial, and temporal information. Environmental variables will include temperature, rainfall, land cover, vegetation condition, and elevation. Additional features such as month, season, and spatial grid information will be derived from the observation dates and geographic coordinates. These variables will be integrated to create a structured dataset suitable for machine-learning analysis.

Random Forest and XGBoost will be developed and compared for bird occurrence prediction. These models are suitable candidates for structured ecological data because they can represent non-linear relationships between bird occurrence and environmental, spatial, and temporal variables. Rather than if one model will perform better, both models will be evaluated under the same experimental conditions, and the final model selection will be based on validation results. Previous research on presence-only species distribution modelling has also shown that factors such as model configuration, background sampling, and parameter tuning can influence predictive performance [4]. Therefore, these factors will be considered when designing the experiments.

The proposed component will generate species-level occurrence probability maps and endemic-bird biodiversity or species-richness maps. In addition to these research outputs, a location-aware feature will identify model-supported candidate observation areas for a selected bird and time. The system will then calculate the distance between the user's selected location and these candidate areas. Since bird occurrence is influenced by many natural and observational factors, the suggested areas will be presented as probability-based predictions rather than guaranteed bird sightings.

The main beneficiaries of this component include bird researchers, ornithologists, conservation organizations, wildlife and forest officers, and field survey teams. Birdwatchers and citizen scientists may also benefit from the location-based observation guidance provided by the system. Within the overall AI-Powered Multimodal Conservation Intelligence Framework for Sri Lankan Endemic Birds, Component 03 is responsible for occurrence and biodiversity prediction. Bird identification from images and audio is handled by Components 01 and 02, while habitat threat assessment and conservation recommendations are addressed by Component 04.

This research is aligned with Sustainable Development Goal 15 – Life on Land, as it aims to support biodiversity monitoring and the conservation of terrestrial species. From a computing perspective, the

component is mainly positioned within Artificial Intelligence and Machine Learning, Data Science, spatio-temporal analysis, and Geographic Information Systems (GIS).

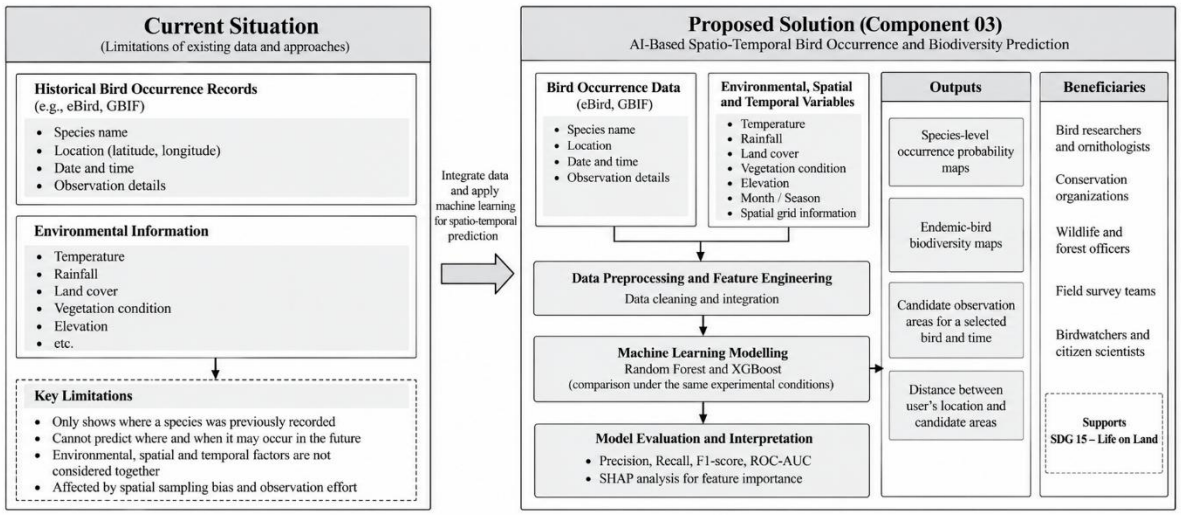


Figure 1 Overview of the Research Problem and Proposed Solution

1.2 Literature Review

Species distribution modelling is a well-established research area that combines species observations with environmental information to estimate where species are likely to occur. Therefore, this research does not consider bird occurrence prediction as a new concept. Instead, existing studies provide a foundation for understanding suitable modelling methods, limitations of community-science data, and opportunities for applying these approaches to Sri Lankan endemic birds.

Fink et al. developed spatial-temporal models using broad-scale bird survey data and demonstrated that species occurrence patterns can change across both space and time [1]. Later eBird research extended this work to model bird distributions and population trends across different periods of the year [2]. These studies support the use of spatial, temporal, and environmental information in Component 03 rather than relying only on historical observation locations.

Johnston et al. examined the use of community-science data such as eBird for species distribution modelling [3]. Their work highlights important issues including uneven sampling, differences in observation effort, and spatial bias. Based on these findings, Component 03 will apply appropriate data filtering and retain useful observation-effort information where available. Missing observations will also not automatically be considered evidence that a bird was absent from a particular location.

Valavi et al. compared several presence-only species distribution modelling methods, including Random Forest and XGBoost [4]. Their study showed that model performance can be influenced by the modelling approach, background sampling, and parameter tuning. This supports the decision to compare Random Forest and XGBoost experimentally and select the model based on validation results rather than assuming beforehand that one model will perform better.

The eBird Status and Trends products demonstrate how bird observations can be combined with environmental, spatial, temporal, and observation-effort information to generate detailed bird distribution products [5]. Component 03 follows a related data-driven principle but focuses specifically on selected Sri Lankan endemic birds. It will generate occurrence probability and endemic-biodiversity information and provide model-supported candidate observation areas with location-based distance guidance.

Sri Lankan biodiversity studies also show the importance of geographic and environmental information for understanding endemic species distributions. This supports the inclusion of variables such as location, vegetation, land cover, elevation, temperature, and rainfall in the proposed approach.

Overall, Component 03 builds on established species-distribution research by developing a Sri Lankan endemic-bird-focused spatio-temporal machine-learning workflow. It integrates occurrence, environmental, spatial, and seasonal information, compares Random Forest and XGBoost, and converts the resulting predictions into occurrence maps, biodiversity information, and practical candidate observation guidance.

Table 1 Comparison of Related Research and Existing Approaches

Study / Approach	**Main Focus**	**Useful Contribution**	**Limitation in This Context**	**Influence on Component 03**
Fink et al. (2010) [1]	Spatio-temporal modelling of broad-scale survey data	Shows species-environment relationships can vary across space and time	Broad-scale framework; not a Sri Lankan endemic-bird application	Supports spatial + temporal feature design
Fink et al. (2020) [2]	Full annual-cycle bird distribution and trends	Demonstrates dynamic bird-distribution modelling from citizen-science data	Large eBird science pipeline differs from this student research scope	Supports time-aware occurrence modelling
Johnston et al. (2021) [3]	Reliable inference from eBird community-science data	Highlights effort, spatial bias and observation-process issues	Guidelines rather than a Sri Lankan endemic prediction system	Influences filtering, effort variables and validation
Valavi et al. (2022) [4]	Benchmark of presence-only SDMs	Compares RF, XGBoost and other methods; stresses fitting/background choices	Not specific to Sri Lankan endemic birds	Supports experimental RF vs XGBoost comparison
eBird Status & Trends [5]	Operational distribution/abundance products	Uses environmental, effort, spatial and temporal information	Not designed as this project's Sri Lankan endemic-bird prototype	Provides an existing-system benchmark
Proposed Component 03	Sri Lankan endemic bird occurrence + biodiversity prediction	Integrates occurrence, environment, space and season; maps + distance guidance	Requires experimental validation	Target research contribution

1.3 Research Gap

Literature makes one thing clear: bird species-distribution modelling, citizen-science modelling and environmental prediction all already exist as research areas. So the gap here isn't the absence of occurrence-prediction methods in general — it's something narrower. Historical Sri Lankan endemic-bird observations, on their own, are not an integrated ML-ready predictive resource that combines occurrence records with environmental, spatial and seasonal information and turns the resulting patterns into both species-level occurrence and endemic-biodiversity outputs suited to local use.

International systems like eBird already provide strong distribution products, and published SDM studies already demonstrate a range of modelling algorithms — but neither directly delivers the specific combination this component investigates: a selected Sri Lankan endemic-bird scope, integration of several local environmental and temporal predictors, an explicit RF-versus-XGBoost experimental comparison, endemic-biodiversity mapping, and location-aware candidate observation guidance, all within one conservation framework.

Research gap statement: There is a need to investigate whether historical occurrence records of selected Sri Lankan endemic birds, combined with spatial, temporal and environmental variables, can support reliable local occurrence and endemic-biodiversity prediction through a validated machine-learning workflow, and whether those predictions can then be translated into useful, map-based candidate observation guidance.

Table 2 Comparison of Related Research and Existing Approaches

Area	Identified Limitation	Proposed Direction
Historical occurrence records	Useful for showing previous observations, but not a complete predictive output	Integrate observations with explanatory variables and train predictive models
Citizen-science bias	Sampling effort and location coverage are uneven	Filter data, retain effort information where available, and use careful validation
Model selection	No algorithm should be assumed best in advance	Compare RF and XGBoost experimentally
Presence-only records	No record does not automatically mean absence	Use a justified non-detection/background/pseudo-absence design
Local usability	Research outputs may remain as technical maps or model files	Provide occurrence/biodiversity maps plus candidate areas and user distance

2. OBJECTIVE

2.1 Main Objective

The main objective of this research is to design, develop, and evaluate a machine-learning-based spatio-temporal framework for predicting the occurrence and biodiversity patterns of selected Sri Lankan endemic bird species by combining historical bird observations with environmental, spatial, and temporal information.

The proposed framework aims to transform historical bird observations into useful predictive information that can support biodiversity monitoring, field survey planning, and the identification of potential bird observation areas.

2.2 Specific Objectives

O1 – Collect and prepare bird occurrence data for selected Sri Lankan endemic species using georeferenced and dated historical records obtained from eBird and GBIF.

O2 – Integrate environmental, spatial, and temporal information with the collected bird observations, including temperature, rainfall, vegetation condition, land cover, elevation, month, season, and spatial-grid information, to prepare a structured dataset **for machine-**learning analysis.

O3 – Develop Random Forest and XGBoost models to predict the occurrence probability of selected endemic bird species using the prepared dataset and an appropriately defined presence and background or non-detection strateg**y.**

O4 – Evaluate and compare the developed models using suitable performance measures and spatially and temporally aware validation methods to identify the better-performing approach.

O5 – Generate spatial prediction outputs from the validated model, including species-level occurrence probability maps and endemic-bird biodiversity or species-richness maps.

O6 – Develop a location-aware prototype feature that identifies model-supported candidate observation areas for a selected bird and time period and calculates the distance between the user's selected location and those areas.

Table 3Mapping of Research Objectives to Expected Outputs

Objective	Research Activity	Expected Output
O1	Bird occurrence data collection and preparation	Clean occurrence dataset
O2	Data integration and feature engineering	ML-ready spatio-temporal environmental dataset
O3	Machine-learning model development	Random Forest and XGBoost candidate models

Objective	Research Activity	Expected Output
O4	Model evaluation and comparison	Validated model comparison and selected approach
O5	Spatial prediction and mapping	Occurrence probability and endemic-biodiversity maps
O6	Location-aware prototype development	Candidate observation areas and distance guidance

3. METHODOLOGY.

3.1 Research Methodology / Research Design

This research will follow a quantitative, experimental, and prototype-based research design. This approach is suitable because the study aims to determine whether historical bird observations, together with environmental, spatial, and temporal information, can be used to predict the occurrence patterns of selected Sri Lankan endemic birds. The research therefore involves data preparation, machine-learning model development, experimental comparison, validation, and the development of practical spatial outputs.

Historical bird occurrence records for selected endemic species will be collected primarily from eBird and GBIF. Relevant information such as species name, geographic coordinates, observation date, and available observation details will be extracted and prepared for analysis. The collected data will be cleaned to remove incomplete, unsuitable, and duplicate records before being used for model development.

The bird occurrence records will then be combined with relevant environmental and geographic variables, including temperature, rainfall, vegetation condition, land cover, and elevation. Spatial and temporal features such as month, season, and spatial-grid information will also be derived from the available records. These variables will be integrated into a structured dataset that represents the environmental, spatial, and seasonal conditions associated with bird observations.

An important consideration in this research is the treatment of species absence. eBird and GBIF records are mainly presence-based, and the absence of a record at a location does not necessarily mean that the species was absent. Therefore, arbitrary absence labels will not be assigned. Where suitable non-detection information is available, it will be considered. Otherwise, an appropriate background or pseudo-absence strategy will be selected based on previous research and documented before final model development.

Random Forest and XGBoost will be developed and compared for bird occurrence prediction. Both models will be trained under the same experimental conditions using the prepared dataset. Their performance will be evaluated using measures such as precision, recall, F1-score, and ROC-AUC, together with spatially and temporally aware validation. The better-performing approach will be selected based on the experimental results rather than being predetermined. SHAP and feature-importance analysis will also be used to examine the influence of different variables on model predictions.

The validated model will be used to generate species-level occurrence probability maps and endemic-bird biodiversity or species-richness maps. The prototype will also identify model-supported candidate observation areas for a selected bird and time period and calculate the distance between the user's selected location and those areas. These outputs will be presented as probability-based predictions rather than guaranteed bird sightings.

This methodology allows the proposed models and outputs to be evaluated systematically while maintaining the main research focus of Component 03: predicting the occurrence and biodiversity patterns of selected Sri Lankan endemic birds using environmental, spatial, and temporal information.

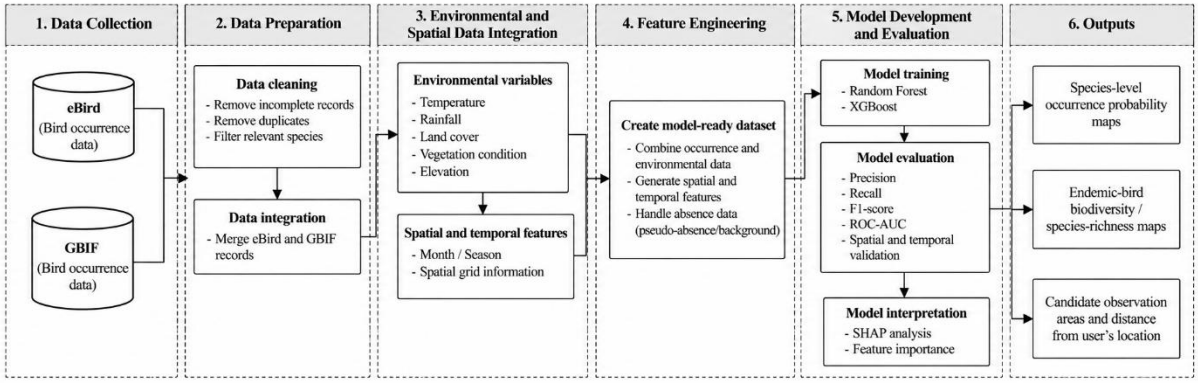


Figure 2 Proposed Component 03 Research Workflow

3.2 Data / Participants / Experimental Inputs

The dataset for this research will be created by combining information from several sources rather than using one ready-made dataset. Bird occurrence records will provide information about where and when birds were observed, while environmental and geographic datasets will describe the conditions associated with those observations. These data will be combined into a structured dataset for machine-learning analysis.

Table 4 Proposed Data Sources and Variables

Data Type	Proposed Source	Example Variables	Purpose
Bird occurrence	eBird and GBIF	Species name, latitude, longitude, observation date; effort/checklist fields where available	Main evidence of where and when birds were observed
Historical weather	Selected historical weather source (e.g., Open-Meteo archive or another approved source)	Temperature, precipitation/rainfall; monthly or seasonal summaries	Represents weather conditions associated with observation time/location
Land cover	MODIS MCD12Q1 or approved equivalent	Land-cover class	Describes whether a location is forest, cropland, urban, water, etc.
Vegetation	MODIS NDVI product such as MOD13Q1, if retained in final design	NDVI / quality information	Represents vegetation condition/greenness
Elevation	SRTM/NASADEM or approved DEM	Elevation in metres	Represents topographic height of the observation location
Derived temporal features	Calculated from observation date	Year, month, season	Allows seasonal patterns to be learned

Data Type	Proposed Source	Example Variables	Purpose
Derived spatial features	Calculated from coordinates	Spatial grid ID and related geographic features	Supports geographic grouping, mapping and spatial validation

The data preparation process will begin by filtering bird observations to **Sri Lanka and the selected endemic species**. Records with invalid coordinates, missing dates, or other important missing information will be removed. Duplicate records will be checked, species names will be standardized, and the number of usable observations available for each species will be reviewed.

Environmental information will then be linked to each bird observation using its **latitude and longitude**. For environmental variables that change over time, the observation date will also be considered when matching the data. Temporal features such as month and season will be derived from the observation date.

A **spatial grid** will be used to divide the study area into smaller geographic cells. Each bird observation will be assigned to a grid based on its coordinates. This will support geographic grouping, biodiversity mapping, and spatial validation. The final grid size will be selected by considering the amount of available bird data and the resolution of the supporting environmental datasets.

Before model training, the available data will be checked for each selected species to determine whether there are enough reliable observations for meaningful analysis. Species with insufficient records may need to be excluded from the main modelling experiment or analyzed separately. Synthetic data may be used only for prototype or interface testing and will not be used to report actual ecological model performance.

3.3 Baseline / Benchmark

To evaluate the proposed AI-Based Endemic Bird Occurrence and Biodiversity Prediction components, the main experimental comparison will be conducted between Random Forest and XGBoost. A historical occurrence map will also be used as a non-machine-learning reference to distinguish historical observation patterns from model-based predictions.

Random Forest will be used as the primary baseline model because it is suitable for structured data and can represent non-linear relationships between bird occurrence and environmental, spatial, and temporal variables. XGBoost will be used as the comparison model because its gradient-boosting approach can capture complex relationships between multiple predictors. The study will not assume that either model is better in advance. Both models will use the same core feature set and comparable training and validation conditions to support a fair experimental comparison.

The historical occurrence map will show locations where selected bird species have previously been recorded using eBird and GBIF data. Unlike the machine-learning models, this reference does not estimate occurrence probability for candidate locations. It therefore provides a practical comparison between displaying past observations and generating predictive occurrence information.

Model performance will be compared using precision, recall, F1-score, ROC-AUC and, where appropriate, PR-AUC, together with spatial and temporal validation. The final model will be selected based on its overall predictive performance, stability, and interpretability.

Table 5 Proposed Baseline and Model Comparison

Approach	Type	Reason for Inclusion	Role
Random Forest	Bagged decision-tree ensemble	Strong baseline for structured, non-linear ecological data	Primary baseline model
XGBoost	Gradient-boosted decision trees	Suitable for modelling complex relationships among multiple predictors	Candidate comparison model
Historical Occurrence Map	Historical observation reference	Shows previously recorded locations but does not predict occurrence in unobserved candidate areas	Non-ML reference output

The baseline comparison will determine whether the proposed machine-learning approach provides useful predictive information beyond the visualization of historical bird observations. Random Forest and XGBoost will be evaluated under comparable experimental conditions, and the selected model will support the generation of occurrence probability maps, endemic-bird biodiversity outputs, and candidate observation areas.

3.4 Evaluation Metrics

The performance of the proposed bird occurrence prediction models will be assessed using multiple evaluation metrics, as no single metric provides a complete measure of predictive performance. This is particularly important when occurrence and non-occurrence/background cases are unevenly distributed. Therefore, precision, recall, F1-score, ROC-AUC, PR-AUC, and the confusion matrix will be considered when evaluating and comparing Random Forest and XGBoost.

Table 6 Evaluation Metrics

Metric	Simple Meaning	Why Relevant
Precision	Proportion of predicted occurrence cases that are correctly classified	Helps assess and control false-positive occurrence predictions
Recall	Proportion of actual occurrence cases correctly identified by the model	Indicates whether the model is missing occurrence cases
F1-score	Harmonic mean of precision and recall	Provides a balanced measure when both false positive and false negatives are important
ROC-AUC	Measures the ability of the model to distinguish occurrence from non-occurrence/background	Provides an overall measure of classification performance

Metric	Simple Meaning	Why Relevant
	cases across different thresholds	
PR-AUC	Evaluates precision and recall across different classification thresholds	Particularly useful when occurrence cases are relatively rare or the classes are imbalanced
Confusion matrix	Summarizes true positives, false positives, true negatives, and false negatives	Provides a clear interpretation of the types of classification errors

No fixed performance threshold will be defined at the proposal stage. Instead, the models will be evaluated under the same experimental conditions, and their performance will be interpreted in relation to the selected baseline, the availability and distribution of data for each species, and the validation strategy.

In addition, spatial and temporal validation will be considered to reduce the risk of overly optimistic results caused by closely related observations appearing in both training and testing data. The final model will therefore be selected based on its overall predictive performance, consistency, and ability to generalize to unseen spatial and temporal conditions.

3.5 Validation Strategy

The validation strategy will assess whether the developed models can generalize beyond the observations used during training. A simple random split may place geographically close or temporally related records in both the training and testing datasets, which could produce overly optimistic performance estimates. Therefore, the study will apply spatially aware validation using geographic groups or grid cells. Temporal validation will also be used where sufficient data are available by separating observations from different time periods, such as years, seasons, or months.

Data quality will be verified before model training by checking coordinates, observation dates, species names, duplicate records, and missing values. Random Forest and XGBoost will then be evaluated using the same core features and comparable validation partitions to ensure a fair model comparison. Performance on held-out data will be assessed using the selected evaluation metrics.

The resulting occurrence probability maps and candidate observation areas will also be reviewed for unrealistic or out-of-scope predictions. Where appropriate, selected outputs will be discussed with the research supervisor or relevant domain experts to support ecological interpretation. This review will complement the quantitative evaluation rather than replace model-based validation.

Table 7 Validation Strategy

Validation Area	What Is Evaluated	Validation Approach	Purpose
Data quality validation	Coordinates, dates, species names, duplicates, and missing values	Automated checks plus manual inspection	Ensure a reliable input dataset
Spatial validation	Generalization across geographic areas	Hold out geographic groups or spatial grid cells	Reduce spatial leakage and assess geographic

Validation Area	What Is Evaluated	Validation Approach	Purpose
			generalization
Temporal validation	Generalization across time periods	Hold out year, season, or month groups where data permit	Assess temporal generalization
Model comparison	Random Forest and XGBoost performance	Use the same core features and comparable validation partitions	Ensure a fair model comparison
Output sanity check	Occurrence maps and candidate observation areas	Review unrealistic or out-of-scope predictions against known ecological context	Improve the reliability of spatial outputs
Domain review	Interpretation of selected model outputs	Review selected predictions with supervisor or domain experts where available	Support ecological interpretation of results

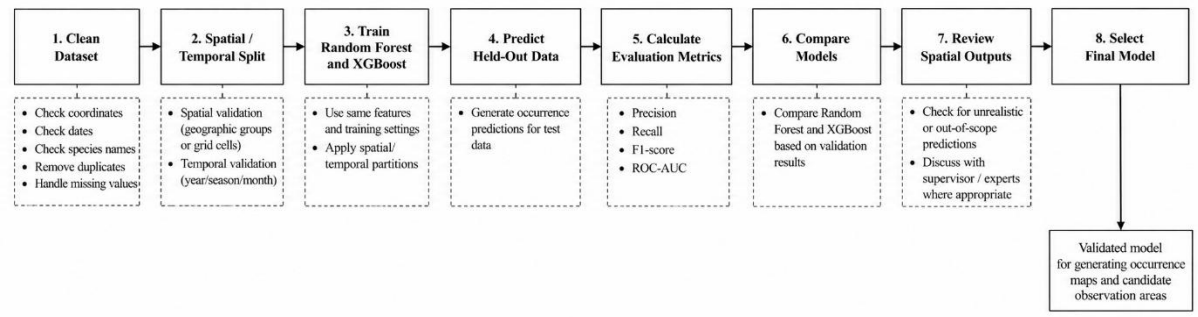


Figure 3 Proposed Validation Process

3.6 Development Approach

The proposed AI-Based Endemic Bird Occurrence and Biodiversity Prediction components will be developed using a modular and iterative prototype-based approach. This allows the main stages of the component to be implemented and evaluated separately before integration into the complete system. Development will cover data preparation and feature engineering, machine-learning model development, model interpretation, and spatial output generation.

Historical bird occurrence records from eBird and GBIF will be cleaned, standardized, and integrated with the selected environmental and geographic variables. Temporal features such as month and season will be derived from observation dates, while latitude and longitude will support spatial processing and grid-based analysis. The resulting structured dataset will provide the common input for model development.

Random Forest and XGBoost will be implemented using the same core feature set and comparable validation conditions. Model performance will be assessed using the defined evaluation metrics together

with spatial and temporal validation. SHAP will be used to examine the contribution of individual predictors to the model outputs and improve the interpretability of the selected model.

The validated model will then be used to generate occurrence probability maps, endemic-bird biodiversity outputs, and candidate observation areas. GIS and mapping tools will support spatial processing and visualization, while location-aware functionality will calculate the distance between the user's location and candidate areas. Development will begin with a limited set of selected endemic bird species to establish the complete workflow, followed by further iterations to improve data quality, model performance, validation, and spatial outputs.

Table 8 Development Technologies and Purpose

Technology	Purpose	Reason for Selection
Python	Data processing and ML implementation	Provides extensive support for machine learning and geospatial analysis
Pandas / NumPy	Data cleaning and feature preparation	Supports efficient processing of structured datasets
GeoPandas / GIS tools	Spatial operations and geospatial data preparation	Supports coordinates, spatial joints, and grid-based operations
Scikit-learn	Random Forest, preprocessing and evaluation	Provides established tools for ML development and evaluation
XGBoost library	XGBoost model implementation	Supports gradient-boosted tree modelling
SHAP	Model explanation	Explains the contribution of predictors to model outputs
QGIS	GIS inspection and data preparation	Supports processing and validation of geographic datasets
Folium / Leaflet or equivalent	Interactive map prototype	Supports occurrence maps and candidate-area visualization

4. HIGH-LEVEL SYSTEM ARCHITECTURE

The overall research project, AI-Powered Multimodal Conservation Intelligence Framework for Sri Lankan Endemic Birds, is designed as an integrated system consisting of four complementary research components. The framework combines image-based bird identification, noise-robust bird-call recognition, endemic bird occurrence and biodiversity prediction, and habitat threat assessment with conservation recommendations. The purpose of architecture is to allow each component to address a distinct research problem while contributing its output to a common conservation intelligence interface.

This proposal focuses on the AI-Based Endemic Bird Occurrence and Biodiversity Prediction component. Its responsibility is to transform historical bird observations together with environmental, spatial, and temporal information into model-supported occurrence and biodiversity predictions. The remaining components are included only to show the integration points and the role of this component within the complete solution.

4.1 High-Level System Architecture

At system level, the image and audio components provide species-identification capabilities, while the occurrence and biodiversity component provides spatial and temporal prediction for selected endemic birds. The habitat threat component separately evaluates habitat risk and provides explainable conservation recommendations. These outputs are presented through the integrated application so that identification, prediction, and conservation information can be accessed within one framework.

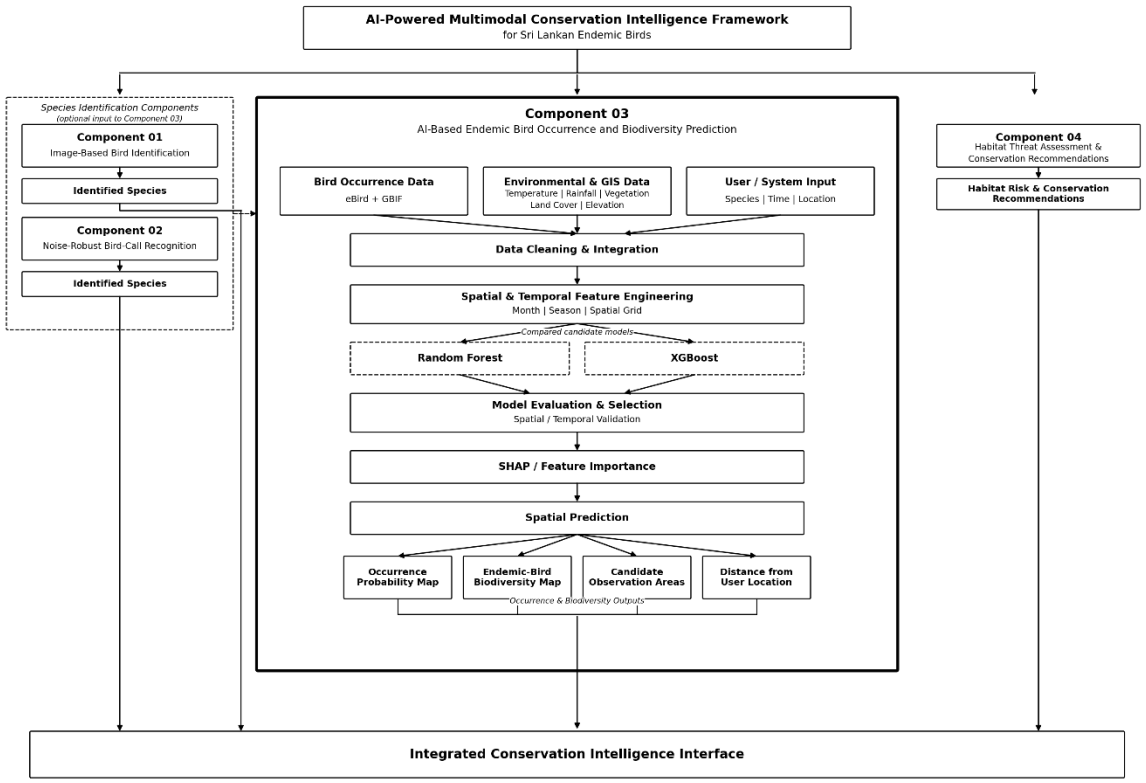


Figure 4 High-Level Architecture of the Overall Research System

4.2 Internal Architecture of AI-Based Endemic Bird Occurrence and Biodiversity Prediction

The internal architecture follows a sequential data-to-prediction workflow. Historical bird occurrence records from eBird and GBIF are combined with selected environmental and GIS variables, including temperature, rainfall, vegetation condition, land cover, and elevation. Data preparation removes invalid or duplicate records and aligns the environmental information with observation locations and dates.

The feature-engineering stage derives temporal features such as month and season and assigns records to appropriate spatial groups or grid cells. The prepared dataset is then used to train and compare Random Forest and XGBoost models. Model evaluation uses held-out data and spatially and temporally aware validation, while SHAP or feature-importance analysis supports interpretation of the main predictors influencing model outputs. The selected model is then applied to geographic areas to produce occurrence-probability maps and endemic-bird biodiversity or species-richness outputs. A location-aware layer identifies model-supported candidate observation areas and calculates their distance from a user-selected location.

4.3 Component Integration and Dependencies

The occurrence and biodiversity prediction component is designed to remain technically independent during model development because its primary inputs are bird occurrence records and environmental, spatial, and temporal datasets. It does not require the image or audio classifiers to train its prediction models. However, integration at application level allows a species identified through the image-based or bird-call recognition component to be passed as a selected species to the occurrence and biodiversity prediction component. The user may also select a supported species directly.

After receiving the species and relevant time information, the component returns occurrence-probability information, biodiversity outputs, candidate observation areas, and distance information to the common interface. The habitat threat assessment component remains a separate analytical responsibility. Although both components may use some environmental datasets, this component does not calculate habitat risk scores or generate conservation recommendations. This separation reduces functional overlap while allowing the overall system to present complementary conservation information.

The planned integration method is a modular, service-based approach using structured inputs and outputs. Common identifiers such as scientific species names, geographic coordinates, and selected time periods will be standardized between components. This allows the components to exchange only the information required for integration without depending on each other's internal model implementation.

4.4 Individual Responsibility Boundary

The individual responsibility boundary defines the work carried out within the AI-Based Endemic Bird Occurrence and Biodiversity Prediction component and separates it from the responsibilities of the other research members. The individual contribution begins with preparation and integration of occurrence and environmental data and ends with validated occurrence, biodiversity, and location-aware prediction outputs. Image recognition, audio recognition, habitat risk assessment, and conservation recommendation generation are outside the scope of this individual research.

Table 9 Individual Responsibility Boundary

Activity / Deliverable	My Responsibility	Shared Responsibility	Other Member
Occurrence data collection, cleaning and preparation	✓		
Environmental and GIS data integration	✓		
Spatial and temporal feature engineering	✓		
Random Forest and XGBoost model development	✓		
Model validation and SHAP / feature-importance analysis	✓		
Occurrence-probability and biodiversity mapping	✓		
Candidate observation area and distance feature	✓		
Common system interface and component integration		✓	Project Team
Image-based bird species identification			Component 01
Noise-robust bird-call recognition			Component 02
Habitat threat assessment and conservation recommendations			Component 04

This boundary keeps the individual research contribution clearly defined while still demonstrating how it supports the complete conservation intelligence framework. The main responsibility of this component ends after generating, validating, explaining, and presenting the occurrence and biodiversity prediction outputs and the associated candidate-area distance information.

5. USER REQUIREMENTS

The intended users of the AI-Based Endemic Bird Occurrence and Biodiversity Prediction components include bird researchers, ornithologists, conservation organizations, field survey teams, and birdwatchers. These users require access to predictive information that supports the understanding of where and when selected Sri Lankan endemic bird species are more likely to occur, together with information on endemic-bird biodiversity patterns across different geographic areas.

At the proposal stage, the requirements are derived from the approved research scope, literature review, identified research gap, proposed methodology, and expected use of the component. The system is expected to provide clear map-based predictions, candidate observation areas, location-aware guidance, and interpretable model outputs. As formal user studies have not yet been completed, the proposal does not claim that the requirements have been validated through interviews or surveys. Relevant stakeholder and domain-expert feedback will be incorporated as supporting evidence when such validation is conducted.

5.1 User Requirement

The user requirements define the main capabilities expected by researchers, conservation users, field teams, and birdwatchers when interacting with the proposed component.

Table 10 User Requirements

ID	User Requirement
UR01	The user should be able to select a supported Sri Lankan endemic bird species for occurrence analysis.
UR02	The user should be able to select a relevant time period, such as a month or season, for prediction.
UR03	The user should be able to view the predicted occurrence probability of a selected species across geographic areas.
UR04	The user should be able to view endemic-bird biodiversity or species-richness patterns across the study area.
UR05	The user should be able to provide or select a location and view model-supported candidate observation areas.
UR06	The user should be able to view the distance between the selected location and identified candidate observation areas.
UR07	The user should be clearly informed that occurrence predictions represent probability-based estimates and do not guarantee bird sightings.
UR08	Research and conservation users should be able to understand the main environmental, spatial, and temporal variables influencing model predictions.

5.2 Functional Requirements

The functional requirements define the operations that the component must perform to transform historical bird occurrence and environmental data into predictive and user-oriented spatial outputs.

Table 11 Functional Requirements

ID	Functional Requirement
FR01	The system shall load, clean, and process support bird occurrence records obtained from the selected data sources.
FR02	The system shall integrate bird occurrence records with relevant environmental and geographic variables.
FR03	The system shall derive temporal features, including month and season, from observation dates.
FR04	The system shall organize observations into appropriate spatial groups or grid cells for analysis and validation.
FR05	The system shall develop and evaluate Random Forest and XGBoost models under comparable experimental conditions.
FR06	The system shall generate species-level occurrence-probability predictions for supported endemic bird species.
FR07	The system shall present predicted occurrence probabilities through geographic map-based outputs.
FR08	The system shall generate endemic-bird biodiversity or species-richness outputs across the study area.
FR09	The system shall allow the user to select a supported bird species and relevant time period for prediction.
FR10	The system shall identify model-supported candidate observation areas based on predicted occurrence probabilities.
FR11	The system shall calculate the distance between the user's selected location and identified candidate observation areas.
FR12	The system shall provide model-interpretation information using SHAP and/or feature-importance analysis.
FR13	The system shall clearly present predictions as probability-based guidance rather than guaranteed bird sightings.

5.3 Non-Functional Requirements

In addition to the required functionality, the component should satisfy appropriate quality requirements related to usability, reliability, reproducibility, performance, explainability, accessibility, scalability, and data integrity. These requirements are particularly important because the component combines multiple ecological and geographic datasets and presents machine-learning predictions to both research-oriented and general users.

Table 12 Non-Functional Requirements

ID	Area	Non-Functional Requirement	Proposed Verification
NFR01	Usability	Users should be able to select a species and time period and interpret the main map-based outputs without requiring specialist GIS knowledge.	Prototype usability review and user testing
NFR02	Reliability	Invalid coordinates, missing required values, and unsupported species should be handled appropriately without producing misleading predictions.	Input validation and error-handling tests
NFR03	Reproducibility	Model experiments should retain the preprocessing steps, model parameters, validation settings, and relevant dataset-version information required to reproduce the results.	Experiment logs and configuration review
NFR04	Performance	Prediction and map requests should be returned within a practical response time for the selected prototype deployment environment.	Response-time testing
NFR05	Explainability	Research-oriented outputs should provide feature-importance or SHAP information to support interpretation of model predictions.	Inspection of model explanation outputs
NFR06	Accessibility	Core interface elements, labels, map information, and controls should remain clear and readable for intended users.	User-interface and accessibility review
NFR07	Scalability	The data and modelling pipeline should support the addition of further endemic bird species without requiring major redesign of the overall workflow.	Testingwith additional species datasets
NFR08	Data Integrity	Data-source information, observation dates, spatial information, and preprocessing records should be retained throughout dataset preparation and model development.	Dataset and preprocessing audit

These requirements establish the expected behavior and quality of the AI-Based Endemic Bird Occurrence and Biodiversity Prediction components. They also provide a basis for prototype development and subsequent system evaluation while keeping the requirements within the defined scope of the individual research component.

6. COMMERCIALIZATION PLAN

Table 13 Commercialization Summary

Area	Proposed Direction
Target Users	Software engineering teams, individual developers and organizations using LLM-based software engineering agents
Problem Addressed	Limited visibility into agent failures, repeated execution problems, weak retrieval, inefficient tool use and brittle planning
Value Proposition	Provides trace-level failure attribution and structured optimization feedback instead of relying only on final task success or failure
Potential Product Form	Developer-focused self-optimizing LLM agent platform or evaluation component integrated with compatible agent workflows
Key User Benefit	Better understanding of where and why agent executions fail and what area may require optimization
HEAE Contribution	Typed trace analysis, prompt/tool/retrieval/planning failure attribution and structured optimization signals
Scalability Direction	Modular architecture and shared schemas allow components to be developed and integrated independently
Deployment Direction	Local-first and self-hosted technology stack selected by the project

6.1 Potential Business Model

The supplied project documents mainly define the **technical and research design** and do not establish a finalized commercial business model, subscription price or revenue forecast. Therefore, these should not be presented as confirmed decisions in the proposal.

A possible future model would be to provide the complete system as a **software development tool for teams or organizations using LLM agents**. Depending on future development, access could potentially be provided through organizational deployment, integration with existing developer workflows, or a service-based model. The project's local-first and self-hosted direction may also be useful for organizations that prefer to keep software engineering traces and code-related information within their own environment.

6.2 User Benefits

The main expected benefit is improved visibility into LLM-agent behaviour. When an execution fails, developers should be able to obtain more useful information than a simple unsuccessful task result. HEAE can identify the relevant execution evidence and attribute the problem to one or more defined failure categories.

The wider system can then use this information for memory and configuration-related decisions. This creates the potential to reduce repeated mistakes and support more reliable agent behavior over multiple executions. However, improvements in development time, monetary cost or productivity should be treated as expected benefits to be evaluated, rather than claimed as proven commercial outcomes at the proposal stage.

7. BUDGET AND JUSTIFICATION

Table 14 Estimated Budget and Resource Justification

Item	Estimated Cost	Justification
Python / pandas / GeoPandas / Rasterio	Free / open source	Data cleaning, processing, spatial joins, raster handling and GIS feature extraction.
Scikit-learn / XGBoost / SHAP	Free / open source	Random Forest and XGBoost model development, evaluation and explainability.
QGIS	Free / open source	Spatial inspection, environmental-data preprocessing and validation of maps.
eBird and GBIF occurrence data	Free / research access	Primary bird occurrence records for selected Sri Lankan endemic species; provider terms and citation requirements must be followed.
Environmental and GIS datasets	Mostly free / open data	Temperature/rainfall, NDVI/vegetation, land cover and elevation variables required for spatial-temporal modelling.
Development computer	Existing resource	Used for preprocessing, feature engineering, tree-based model training, evaluation and prototype development.
Cloud compute	Optional - TBD	Additional processing or model experimentation if the final dataset becomes too large for local computing resources.
Cloud / project storage	Optional - TBD	Backup and storage of occurrence data, environmental layers, processed datasets, models and generated outputs.
Web hosting / deployment service	TBD	Deployment of the research prototype, map visualizations and user-facing prediction interface.
Internet and data transfer	Existing / TBD	Downloading biodiversity and environmental datasets, accessing research services and uploading project outputs.
External supervisor / domain-expert consultation	Existing project support	Supports review of species scope, dataset suitability and ecological interpretation of research outputs.
Field or output validation support	Optional - TBD	Supports practical review of predicted occurrence areas and biodiversity outputs where

Item	Estimated Cost	Justification
		feasible; predictions remain probabilistic.
Mapping / distance service	Free tier / TBD	Supports location-aware candidate observation guidance and distance calculation in the prototype.

8. WORK BREAKDOWN STRUCTURE (WBS)

The Work Breakdown Structure divides the research and development activities of the AI-Based Endemic Bird Occurrence and Biodiversity Prediction component into manageable stages. The work begins with literature review and research-gap analysis, followed by data collection, preprocessing, environmental and spatial data integration, feature engineering, machine-learning model development, validation, spatial-output generation, and prototype integration.

The WBS focuses specifically on the individual responsibilities of this research. Development of image-based bird identification, noise-robust bird-call recognition, and habitat-threat assessment remain outside the scope of this component. Integration activities are limited to connecting the validated occurrence and biodiversity outputs with the overall conservation intelligence system.

Table 15 Work Breakdown Structure

WBS ID	Main Activity	Sub-Activity / Task	Expected Output
1.0	Research and Analysis	Review literature related to bird occurrence prediction, species-distribution modelling, biodiversity analysis, and spatio-temporal modelling	Literature review and identified research gap
1.1	Research and Analysis	Analyse existing research approaches and related biodiversity systems	Comparison of existing approaches
1.2	Research and Analysis	Define research objectives, scope, and individual responsibility boundary	Finalized objectives and research scope
2.0	Requirements and Design	Define user, functional, and non-functional requirements	Requirements specification
2.1	Requirements and Design	Design the component architecture and data-processing workflow	Component architecture and workflow
2.2	Requirements and Design	Define data requirements, input variables, and expected outputs	Data and feature specification
3.0	Data Preparation	Collect bird occurrence records from eBird and GBIF	Bird occurrence dataset
3.1	Data Preparation	Collect relevant environmental and geographic data	Environmental and GIS datasets

WBS ID	Main Activity	Sub-Activity / Task	Expected Output
3.2	Data Preparation	Clean, standardize, and integrate the collected datasets	Integrated research dataset
4.0	Feature Engineering	Derive temporal features such as month and season	Temporal features
4.1	Feature Engineering	Create spatial-grid and geographic features	Spatial features
4.2	Feature Engineering	Prepare the final model-ready dataset	Model-ready dataset
5.0	Model Development	Develop Random Forest and XGBoost models	Trained prediction models
5.1	Model Development	Tune and compare both models	Model comparison results
6.0	Validation	Evaluate models using selected performance metrics	Model evaluation results
6.1	Validation	Perform spatial and temporal validation	Generalization results
6.2	Model Interpretation	Apply SHAP and feature-importance analysis	Model interpretation results
7.0	Spatial Output Development	Generate occurrence probability maps	Species occurrence maps
7.1	Spatial Output Development	Generate endemic-bird biodiversity maps	Biodiversity maps
7.2	Spatial Output Development	Identify candidate observation areas and calculate distance	Location-aware guidance
8.0	Prototype and Integration	Develop and integrate the component interface	Functional prototype
8.1	Prototype and Integration	Conduct component and integration testing	Tested component

WBS ID	Main Activity	Sub-Activity / Task	Expected Output
9.0	Documentation	Maintain research and experimental documentation	Research records
9.1	Documentation	Prepare final report, presentation, and demonstration	Final research deliverables

9. GANTT CHART

Gantt Chart – Component 03

AI-Based Spatio-Temporal Occurrence and Biodiversity Prediction for Sri Lankan Endemic Birds

Phase	Research Activity	Jun 2026	Jul 2026	Aug 2026	Sep 2026	Oct 2026	Nov 2026	Dec 2026	Jan 2027	Feb 2027
1. Literature Work	Literature review on species distribution modelling and biodiversity prediction									
	Review of datasets (eBird, GBIF, environmental data)									
2. Research Design	Problem definition and research gap analysis									
	Methodology design (features, modelling approach)									
	Experimental design and evaluation plan									
3. Data Preparation	Collect and preprocess occurrence data (eBird, GBIF)									
	Integrate environmental variables (weather, land cover, elevation, etc.)									
	Feature engineering and spatial-temporal dataset creation									
4. Experiments / Development	Model development (e.g., Random Forest, XGBoost)									
	Model training and hyperparameter tuning									
	Validation setup (spatial and temporal validation)									
5. Analysis	Model evaluation and performance comparison									
	Biodiversity analysis and candidate area identification									
6. Prototype Work	Develop component interface and map visualizations									
7. Testing	System testing and result validation									
8. Integration	Integration with overall research framework									
9. Evaluation	Final evaluation and result interpretation									
10. Report Preparation	Write final report and prepare presentation									

10. REFERENCES

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11. APPENDICES

11.1 Appendix 1: Proposal Supporting Evidence

A1.1 Preliminary Dataset Evidence

- Bird occurrence data sources

  eBird — bird observation data source
  GBIF — biodiversity occurrence data source
  WorldClim — climate and environmental data source
  MODIS — vegetation and land-cover data source
  SRTM — elevation data source

A1.2 External Supervisor Dataset Approval and Dataset Evidence

The project has received dataset approval and supporting dataset material from the external supervisor. The approved material will be stored in the project Google Drive and provided as supporting evidence for the proposed research. This allows the approval and dataset evidence to be reviewed without embedding large data files directly in the proposal.

Dataset Approval and Dataset Evidence: Research_Dataset External approval Letter

A1.3 Prototype Evidence

The project prototype provides supporting evidence of the proposed system design and demonstrates how the main research components and user-facing outputs are intended to be presented through the integrated application.

Prototype Link: System prototype

A1.3 Prototype Evidence

Table 16 Literature Comparison Matrix

Study	Focus	Useful Contribution	Relevant Limitation	Decision / Influence
Fink et al. [1]	Spatio-temporal distribution modelling	Space/time-varying modelling	Not Sri Lankan endemic scope	Supports spatio-temporal design
Fink et al. [2]	Annual-cycle bird distribution	Dynamic bird predictions	Large operational eBird framework	Supports time-aware prediction
Johnston et al. [3]	Citizen-science analytical guidance	Bias/effort-aware modelling	Guidance, not local product	Supports filtering and validation
Valavi et al. [4]	Presence-only model benchmark	RF/XGBoost and background design	Multi-region benchmark, not Sri Lanka	Supports model comparison and careful background sampling
Component	Sri Lankan	Local integrated	Proposed; requires	Research

Study	Focus	Useful Contribution	Relevant Limitation	Decision / Influence
03	endemic occurrence + biodiversity	ML + map guidance	validation	contribution

11.2 Appendix 2 — Proposal Evidence and Decision Log

The following log records key research and design decisions made during proposal development. Dates, collected evidence, and supervisor-discussion details should reflect the actual project records. The log should be updated throughout the research when important decisions are made regarding data, modelling, validation, system design, or implementation.

Table 17 Evidence and Decision Log

Date	Claim / Problem	Evidence Collected	Alternatives Considered	Decision Made	Supervisor Discussion
2nd August 2026	The research team discussed whether the project should focus on bird species worldwide or Sri Lankan endemic bird species.	The team considered a worldwide bird scope and a Sri Lankan endemic-bird-specific scope. The research focus, feasibility, conservation relevance, and project scope were discussed with the supervisor.	Worldwide bird species; Sri Lankan bird species; Sri Lankan endemic bird species.	The research was scoped specifically to Sri Lankan endemic bird species to maintain a focused, locally relevant, and manageable research scope.	The alternatives were discussed with the supervisor, and the project focus was refined to Sri Lankan endemic bird species.
15th August 2026	The research team needed to determine whether the externally obtained bird dataset was suitable for the proposed project.	The externally obtained dataset was presented to the supervisor and reviewed against the project's selected species scope and research requirements.	Use the external dataset; rely only on publicly available datasets; combine external and appropriate public data sources where required.	The external dataset was accepted as supporting data for the project, with each component preparing and preprocessing the relevant data according to its individual research requirements.	The dataset was presented and discussed with the supervisor, including its relevance and suitability for the proposed research components.

11.3 Appendix 3: AI Use Disclosure

Table 18 AI Use Disclosure

AI Tool	Purpose of Use	What Was Generated/Assisted	How Output Was Verified	Final Student Contribution
ChatGPT	Proposal writing and organization support	Assisted with organizing proposal sections, improving academic wording, simplifying technical explanations, structuring tables, and identifying areas requiring further evidence or clarification.	Outputs were reviewed against the approved Topic Assessment Form, proposal template, supervisor feedback, dataset documentation, research scope, and relevant academic literature. References and technical claims were checked before inclusion.	The student finalized the research problem, objectives, methodology, dataset requirements, Random Forest and XGBoost approach, evaluation strategy, component scope, research decisions, references, and interpretation of results.
Claude	Writing support and non-architectural visual assistance	Assisted with refining selected written content and generating supporting visual material for presentation or documentation purposes. It was not used to create the system architecture or other required technical architecture diagrams.	Generated material was manually reviewed for consistency with the approved research scope and compared with the student's actual research content before use. Visual material was checked to ensure that it did not misrepresent the proposed methodology or research outputs.	The student selected, reviewed, corrected, and approved the final content and visual material. All technical diagrams, architecture decisions, research methodology, data preparation, experiments, and research conclusions remain the student's responsibility.

12. CONCLUSION

This work proposes the AI-Based Spatio-Temporal Endemic Bird Occurrence and Biodiversity Prediction component for the AI-Powered Multimodal Conservation Intelligence Framework for Sri Lankan Endemic Birds. The component combines historical bird observations with environmental, spatial, and temporal information to predict where and when selected endemic bird species are more likely to occur.

Bird occurrence records from eBird and GBIF will be integrated with variables such as temperature, rainfall, vegetation, land cover, elevation, location, month, and season. Random Forest and XGBoost will be compared to estimate occurrence probabilities, while SHAP will be used to interpret the influence of key features on model predictions.

The main outputs will include occurrence probability maps, endemic-bird biodiversity maps, candidate observation areas, and distance-based guidance from the user's location. These outputs are intended to provide predictive information rather than guarantee bird sightings.

The main contribution of this component is to move beyond the display of historical occurrence records by providing model-supported spatial and temporal predictions that can support biodiversity monitoring, conservation research, and bird observation planning. Model performance will be evaluated using Precision, Recall, F1-score, ROC-AUC, PR-AUC, and spatially aware validation, with temporal validation applied where sufficient data are available.